

Sutton and Barton Exercises Chapter 12

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1 Chapter 12

Exercise 12.1 Just as the return can be written recursively in terms of the first reward and itself one-step later (3.9), so can the λ -return. Derive the analogous recursive relationship from (12.2) and (12.1).

$$\begin{aligned} G_t^\lambda &= (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} G_t^{(n)} \\ &= (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} \left[R_{t+1} + \gamma G_{t+1}^{(n-1)} \right] \\ &= (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} R_{t+1} + (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} \gamma G_{t+1}^{(n-1)} \\ &= R_{t+1} + \gamma (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} G_{t+1}^{(n-1)} \\ &= R_{t+1} + \gamma (1 - \lambda) \sum_{k=0}^{\infty} \lambda^k G_{t+1}^{(k)} \\ &= R_{t+1} + \gamma (1 - \lambda) \left[G_{t+1}^{(0)} + \sum_{k=1}^{\infty} \lambda^k G_{t+1}^{(k)} \right] \\ &= R_{t+1} + \gamma (1 - \lambda) V(S_{t+1}) + \gamma (1 - \lambda) \sum_{k=1}^{\infty} \lambda^k G_{t+1}^{(k)} \\ &= R_{t+1} + \gamma (1 - \lambda) V(S_{t+1}) + \gamma \lambda \left[(1 - \lambda) \sum_{k=1}^{\infty} \lambda^{k-1} G_{t+1}^{(k)} \right] \\ &= R_{t+1} + \gamma (1 - \lambda) V(S_{t+1}) + \gamma \lambda G_{t+1}^\lambda \\ &= R_{t+1} + \gamma \left[(1 - \lambda) V(S_{t+1}) + \lambda G_{t+1}^\lambda \right]. \end{aligned}$$

Exercise 12.2 The parameter λ characterizes how fast the exponential weighting in Figure 12.2 falls off, and thus how far into the future the λ -return algorithm looks in determining its update. But a rate factor such as λ is sometimes an awkward way of characterizing the speed of the decay. For some purposes it is better to specify a time constant, or half-life. What is the equation relating λ and the half-life τ_λ , the time by which the weighting sequence will have fallen to half of its initial value?

Solution:

The weight at step n is $(1-\lambda)\lambda^{n-1}$. The initial value at $n = 1$ is $(1-\lambda)$. We want τ_λ such that the weight has fallen to half:

$$(1-\lambda)\lambda^{\tau_\lambda-1} = \frac{1}{2}(1-\lambda)$$

Cancel $(1-\lambda)$:

$$\lambda^{\tau_\lambda-1} = \frac{1}{2}$$

Take the natural log of both sides:

$$(\tau_\lambda - 1) \ln \lambda = -\ln 2$$

Solve for τ_λ :

$$\tau_\lambda = 1 - \frac{\ln 2}{\ln \lambda}$$

Exercise 12.3 Some insight into how $\text{TD}(\lambda)$ can closely approximate the off-line λ -return algorithm can be gained by seeing that the latter's error term (in brackets in (12.4)) can be written as the sum of TD errors (12.6) for a single fixed $\mathbf{w}$. Show this, following the pattern of (6.6), and using the recursive relationship for the λ -return you obtained in Exercise 12.1.

Solution:

Recall from (6.6) that for fixed $\mathbf{w}$:

$$G_t - \hat{v}(S_t, \mathbf{w}) = \sum_{k=t}^{T-1} \gamma^{k-t} \delta_k$$

We want the analogous result for the λ -return error. From Exercise 12.1:

$$G_t^\lambda = R_{t+1} + \gamma [(1-\lambda)\hat{v}(S_{t+1}, \mathbf{w}) + \lambda G_{t+1}^\lambda]$$

Subtract $\hat{v}(S_t, \mathbf{w})$ from both sides:

$$G_t^\lambda - \hat{v}(S_t, \mathbf{w}) = R_{t+1} + \gamma(1-\lambda)\hat{v}(S_{t+1}, \mathbf{w}) + \gamma\lambda G_{t+1}^\lambda - \hat{v}(S_t, \mathbf{w})$$

Add and subtract $\gamma \hat{v}(S_{t+1}, \mathbf{w})$:

$$= \underbrace{R_{t+1} + \gamma \hat{v}(S_{t+1}, \mathbf{w}) - \hat{v}(S_t, \mathbf{w})}_{\delta_t} + \gamma \lambda (G_{t+1}^\lambda - \hat{v}(S_{t+1}, \mathbf{w}))$$

Expanding recursively:

$$= \delta_t + \gamma \lambda \delta_{t+1} + (\gamma \lambda)^2 \delta_{t+2} + \dots$$

$$G_t^\lambda - \hat{v}(S_t, \mathbf{w}) = \sum_{k=t}^{T-1} (\gamma \lambda)^{k-t} \delta_k$$

Exercise 12.4 Use your result from the preceding exercise to show that, if the weight updates over an episode were computed on each step but not actually used to change the weights ($\mathbf{w}$ remained fixed), then the sum of TD(λ)'s weight updates would be the same as the sum of the off-line λ -return algorithm's updates.

As $\lambda \rightarrow 0$, $\tau_\lambda \rightarrow 1$ (weights die after one step). As $\lambda \rightarrow 1$, $\tau_\lambda \rightarrow \infty$ (weights decay very slowly, approaching Monte Carlo).

Solution:

The TD(λ) update at each step with eligibility trace $\mathbf{z}_t = \sum_{k=0}^t (\gamma \lambda)^{t-k} \nabla \hat{v}(S_k, \mathbf{w})$ is:

$$\Delta \mathbf{w}_t = \alpha \delta_t \mathbf{z}_t$$

The sum of all updates over the episode is:

$$\sum_{t=0}^{T-1} \alpha \delta_t \mathbf{z}_t = \alpha \sum_{t=0}^{T-1} \delta_t \sum_{k=0}^t (\gamma \lambda)^{t-k} \nabla \hat{v}(S_k, \mathbf{w})$$

Swap the order of summation:

$$= \alpha \sum_{k=0}^{T-1} \nabla \hat{v}(S_k, \mathbf{w}) \sum_{t=k}^{T-1} (\gamma \lambda)^{t-k} \delta_t$$

Apply the result from Exercise 12.3:

$$\sum_{t=k}^{T-1} (\gamma \lambda)^{t-k} \delta_t = G_k^\lambda - \hat{v}(S_k, \mathbf{w})$$

Substituting:

$$\sum_{t=0}^{T-1} \alpha \delta_t \mathbf{z}_t = \alpha \sum_{k=0}^{T-1} \nabla \hat{v}(S_k, \mathbf{w}) (G_k^\lambda - \hat{v}(S_k, \mathbf{w}))$$

Which is exactly the sum of the offline λ -return algorithm's updates — one gradient step per timestep toward the λ -return target.

Exercise 12.5 Several times in this book we have established that returns can be written as sums of TD errors if the value function is held constant. Why is (12.10) another instance of this? Prove (12.10).

Explanation:

Equation (12.10) is another instance of the sum-of-TD-errors pattern because each δ'_i treats the weights as locally fixed for that one step — $\mathbf{w}_{i-1}$ for the current state and $\mathbf{w}_i$ for the next state. The telescoping cancellation still holds step-by-step, just with a slightly different δ' that accounts for the weight update between steps.

Proof of (12.10):

We want to show:

$$G_{t:t+k}^\lambda = \hat{v}(S_t, \mathbf{w}_{t-1}) + \sum_{i=t}^{t+k-1} (\gamma\lambda)^{i-t} \delta'_i$$

where $\delta'_i = R_{i+1} + \gamma\hat{v}(S_{i+1}, \mathbf{w}_i) - \hat{v}(S_i, \mathbf{w}_{i-1})$.

Expand the sum by writing out each term:

$$\sum_{i=t}^{t+k-1} (\gamma\lambda)^{i-t} \delta'_i = \delta'_t + \gamma\lambda\delta'_{t+1} + (\gamma\lambda)^2\delta'_{t+2} + \cdots + (\gamma\lambda)^{k-1}\delta'_{t+k-1}$$

Substituting $\delta'_i = R_{i+1} + \gamma\hat{v}(S_{i+1}, \mathbf{w}_i) - \hat{v}(S_i, \mathbf{w}_{i-1})$:

$$\begin{aligned} &= [R_{t+1} + \gamma\hat{v}(S_{t+1}, \mathbf{w}_t) - \hat{v}(S_t, \mathbf{w}_{t-1})] \\ &\quad + \gamma\lambda[R_{t+2} + \gamma\hat{v}(S_{t+2}, \mathbf{w}_{t+1}) - \hat{v}(S_{t+1}, \mathbf{w}_t)] \\ &\quad + (\gamma\lambda)^2[R_{t+3} + \gamma\hat{v}(S_{t+3}, \mathbf{w}_{t+2}) - \hat{v}(S_{t+2}, \mathbf{w}_{t+1})] \\ &\quad + \cdots + (\gamma\lambda)^{k-1}[R_{t+k} + \gamma\hat{v}(S_{t+k}, \mathbf{w}_{t+k-1}) - \hat{v}(S_{t+k-1}, \mathbf{w}_{t+k-2})] \end{aligned}$$

Notice the telescoping: the $+\gamma\hat{v}(S_{i+1}, \mathbf{w}_i)$ term from step i cancels with the $-\hat{v}(S_{i+1}, \mathbf{w}_i)$ term from step $i+1$ (after absorbing the $\gamma\lambda$ factor). What remains is:

$$= -\hat{v}(S_t, \mathbf{w}_{t-1}) + \sum_{i=t}^{t+k-1} (\gamma\lambda)^{i-t} R_{i+1} + (\gamma\lambda)^{k-1} \cdot \gamma\hat{v}(S_{t+k}, \mathbf{w}_{t+k-1})$$

Rearranging:

$$\hat{v}(S_t, \mathbf{w}_{t-1}) + \sum_{i=t}^{t+k-1} (\gamma\lambda)^{i-t} \delta'_i = \sum_{i=t}^{t+k-1} (\gamma\lambda)^{i-t} R_{i+1} + \gamma^k \lambda^{k-1} \hat{v}(S_{t+k}, \mathbf{w}_{t+k-1})$$

Which matches the definition of $G_{t:t+k}^\lambda$ — a k -step λ -return bootstrapping at S_{t+k} . $\square$

Exercise 12.6 Modify the pseudocode for Sarsa(λ) to use dutch traces (12.11) without the other distinctive features of a true online algorithm. Assume linear function approximation and binary features.

Solution:

Algorithm 1 Sarsa(λ) with Dutch traces and binary features

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1: Input: a function  $\mathcal{F}(s, a)$  returning the set of (indices of) active features
   for  $s, a$ 
2: Input: a policy  $\pi$ 
3: Algorithm parameters: step size  $\alpha > 0$ , trace decay rate  $\lambda \in [0, 1]$ , small
    $\varepsilon > 0$ 
4: Initialize:  $\mathbf{w} = (w_1, \dots, w_d)^\top \in \mathbb{R}^d$  (e.g.,  $\mathbf{w} = \mathbf{0}$ ),  $\mathbf{z} = (z_1, \dots, z_d)^\top \in \mathbb{R}^d$ 
5: for each episode do
6:   Initialize  $S$ 
7:   Choose  $A \sim \pi(\cdot|S)$  or  $\varepsilon$ -greedy according to  $\hat{q}(S, \cdot, \mathbf{w})$ 
8:    $\mathbf{z} \leftarrow \mathbf{0}$ 
9:   for each step of episode do
10:    Take action  $A$ , observe  $R, S'$ 
11:     $\delta \leftarrow R$ 
12:    for  $i$  in  $\mathcal{F}(S, A)$  do
13:       $\delta \leftarrow \delta - w_i$ 
14:       $z_i \leftarrow (1 - \alpha z_i)z_i + 1$  ▷ Dutch trace update
15:    end for
16:    if  $S'$  is terminal then
17:       $\mathbf{w} \leftarrow \mathbf{w} + \alpha \delta \mathbf{z}$ 
18:      Go to next episode
19:    end if
20:    Choose  $A' \sim \pi(\cdot|S')$  or  $\varepsilon$ -greedy according to  $\hat{q}(S', \cdot, \mathbf{w})$ 
21:    for  $i$  in  $\mathcal{F}(S', A')$  do
22:       $\delta \leftarrow \delta + \gamma w_i$ 
23:    end for
24:     $\mathbf{w} \leftarrow \mathbf{w} + \alpha \delta \mathbf{z}$ 
25:     $\mathbf{z} \leftarrow \gamma \lambda \mathbf{z}$ 
26:     $S \leftarrow S'; A \leftarrow A'$ 
27:  end for
28: end for

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Exercise 12.7: Generalize the three recursive equations (12.18, 12.19, 12.20) to their truncated versions, defining $G_{t:h}^{\lambda s}$ and $G_{t:h}^{\lambda a}$.

$$\begin{aligned}
G_{t:h} &= \begin{cases} R_{t+1} + \gamma G_{t+1:h}, & t < h \\ 0, & t = h \end{cases} \\
G_{t:h}^\lambda &= \begin{cases} R_{t+1} + \gamma [(1 - \lambda)V(S_{t+1}) + \lambda G_{t+1:h}^\lambda], & t < h \\ V(S_h), & t = h \end{cases} \\
v_{t:h}(S_t) &= \mathbb{E} \left[\begin{cases} R_{t+1} + \gamma v_{t+1:h}(S_{t+1}), & t < h \\ 0, & t = h \end{cases} \right]
\end{aligned}$$

Exercise 12.8: Prove that (12.24) becomes exact if the value function does not change. To save writing, consider the case of $t = 0$, and use the notation $V_k \doteq \hat{v}(S_k, \mathbf{w})$.

Solution: Begin by expanding $G_0^{\lambda s}$ using (12.22):

$$G_0^{\lambda s} = \rho_0 \left(R_1 + \gamma_1 ((1 - \lambda_1)V_1 + \lambda_1 G_1^{\lambda s}) \right) + (1 - \rho_0)V_0 \quad (1)$$

Recall the state-based TD error (12.23):

$$\delta_k^s \doteq R_{k+1} + \gamma_{k+1}V_{k+1} - V_k \quad (2)$$

We can rearrange to get:

$$R_{k+1} + \gamma_{k+1}V_{k+1} = \delta_k^s + V_k \quad (3)$$

Now substitute back into the expansion of $G_0^{\lambda s}$ and expand $G_1^{\lambda s}$ recursively. At each step k , the $(1 - \rho_k)V_k$ terms and the bootstrapped V_k terms from δ_k^s telescope cleanly because the value function is fixed. After full recursive expansion, all intermediate V_k terms cancel except V_0 , leaving:

$$G_0^{\lambda s} = V_0 + \rho_0 \sum_{k=0}^{\infty} \delta_k^s \prod_{i=1}^k \gamma_i \lambda_i \rho_i \quad (4)$$

which is exactly (12.24) for $t = 0$. The approximation in (12.24) arises only when the value function changes between updates (i.e. $\mathbf{w}$ is updated), so that $\hat{v}(S_k, \mathbf{w}_t) \neq \hat{v}(S_k, \mathbf{w}_k)$. When $\mathbf{w}$ is fixed, the telescoping is exact. $\square$

Exercise 12.9: The truncated version of the general off-policy return is denoted $G_{t:h}^{\lambda s}$. Guess the correct equation, based on (12.24).

Solution: By analogy with (12.24), we truncate the infinite sum at horizon h :

$$G_{t:h}^{\lambda s} \doteq \hat{v}(S_t, \mathbf{w}_t) + \rho_t \sum_{k=t}^{h-1} \delta_k^s \prod_{i=t+1}^k \gamma_i \lambda_i \rho_i \quad (5)$$

The structure is identical to (12.24) except the sum runs only to $h-1$ rather than ∞ . Bootstrapping beyond h is handled implicitly through the δ_k^s terms, which already contain $\hat{v}(S_{k+1}, \mathbf{w})$. As with (12.24), this becomes exact when the value function does not change. $\square$ `latexarticle amsmath amssymb`

Exercise 12.10

Question

Prove that (12.27) becomes exact if the value function does not change. To save writing, consider the case of $t = 0$, and use the notation $Q_k = \hat{q}(S_k, A_k, \mathbf{w})$.

Answer

We wish to show that the eligibility-trace-based update is exact when $\mathbf{w}$ is fixed. We expand $G_0^{\lambda a}$ term by term by unrolling the recursive definition.

The expected TD error at step k is:

$$\delta_k^s = R_{k+1} + \gamma \bar{V}(S_{k+1}) - Q_k$$

where $\bar{V}(S_k) = \sum_a \pi(a|S_k) \hat{q}(S_k, a, \mathbf{w})$ is the expected value under the policy.

At $k = 0$: The first term of the unrolled return is:

$$Q_0 + \delta_0^s = Q_0 + R_1 + \gamma \bar{V}(S_1) - Q_0 = R_1 + \gamma \bar{V}(S_1)$$

At $k = 1$: Adding the next term $\gamma \lambda \delta_1^s$:

$$R_1 + \gamma \bar{V}(S_1) + \gamma \lambda \delta_1^s = R_1 + \gamma \bar{V}(S_1) + \gamma \lambda (R_2 + \gamma \bar{V}(S_2) - Q_1)$$

At $k = 2$: Adding the next term $(\gamma \lambda)^2 \delta_2^s$:

$$R_1 + \gamma \bar{V}(S_1) + \gamma \lambda (R_2 + \gamma \bar{V}(S_2) - Q_1) + (\gamma \lambda)^2 (R_3 + \gamma \bar{V}(S_3) - Q_2)$$

The general pattern is now clear. At each step $k \geq 1$, the term $-Q_k$ from δ_k^s and the term $\gamma \bar{V}(S_k)$ from δ_{k-1}^s appear together. When $\mathbf{w}$ is fixed, we have:

$$\bar{V}(S_k) = \sum_a \pi(a|S_k) \hat{q}(S_k, a, \mathbf{w})$$

and since the action A_k is chosen according to π , the expected value satisfies $\gamma \bar{V}(S_k) = \gamma Q_k$, so these terms cancel exactly:

$$\gamma \bar{V}(S_k) - \gamma Q_k = 0 \quad \text{for all } k \geq 1$$

This telescoping cancellation propagates through all intermediate steps, leaving only the discounted reward stream and the terminal bootstrap term, which is precisely the definition of the λ -return $G_0^{\lambda a}$. Therefore (12.27) becomes exact when the value function does not change. $\square$

Exercise 12.11

section*Exercise 12.11

Question

The truncated version of the general off-policy return is denoted $G_{t:h}^{\lambda a}$. Guess the correct equation for it, based on (12.27).

Answer

By analogy with the state case, the truncated action-value λ -return is obtained by simply capping the summation at horizon h instead of running to infinity. Recall the full off-policy return (12.27):

$$G_t^{\lambda a} \approx \hat{q}(S_t, A_t, \mathbf{w}_t) + \sum_{k=t}^{\infty} \delta_k^s \prod_{i=t+1}^k \gamma_i \lambda_i \rho_i$$

where the expected TD error is:

$$\delta_k^s = R_{k+1} + \gamma_{k+1} \bar{V}(S_{k+1}) - \hat{q}(S_k, A_k, \mathbf{w}_k)$$

and $\bar{V}(S_k) = \sum_a \pi(a|S_k) \hat{q}(S_k, a, \mathbf{w})$ is the expected value under the target policy, and $\rho_i = \frac{\pi(A_i|S_i)}{b(A_i|S_i)}$ are the importance sampling ratios for off-policy correction.

The truncated version is then:

$$G_{t:h}^{\lambda a} \doteq \hat{q}(S_t, A_t, \mathbf{w}_t) + \sum_{k=t}^{h-1} \delta_k^s \prod_{i=t+1}^k \gamma_i \lambda_i \rho_i$$

In the on-policy case $\rho_i = 1$ for all i , so the importance sampling ratios disappear and we recover the simpler on-policy form. $\square$

Exercise 12.12

Question

Show in detail the steps outlined above for deriving (12.29) from (12.27). Start with the update (12.15), substitute $G_t^{\lambda^a}$ from (12.26) for G_t^λ , then follow similar steps as led to (12.25).

Answer

We begin with the semi-gradient update (12.15) for action values:

$$\mathbf{w}_{t+1} \doteq \mathbf{w}_t + \alpha [G_t^{\lambda^a} - \hat{q}(S_t, A_t, \mathbf{w}_t)] \nabla \hat{q}(S_t, A_t, \mathbf{w}_t)$$

From (12.27), we have:

$$G_t^{\lambda^a} - \hat{q}(S_t, A_t, \mathbf{w}_t) = \sum_{k=t}^{\infty} \delta_k^s \prod_{i=t+1}^k \gamma_i \lambda_i \rho_i$$

where the expected action-based TD error is:

$$\delta_k^s = R_{k+1} + \gamma_{k+1} \bar{V}(S_{k+1}) - \hat{q}(S_k, A_k, \mathbf{w}_k)$$

Substituting into the update and summing over all timesteps:

$$\sum_{t=0}^{\infty} (\mathbf{w}_{t+1} - \mathbf{w}_t) \approx \sum_{t=0}^{\infty} \sum_{k=t}^{\infty} \alpha \nabla \hat{q}(S_t, A_t, \mathbf{w}_t) \delta_k^s \prod_{i=t+1}^k \gamma_i \lambda_i \rho_i$$

Swapping the order of summation using the rule $\sum_{t=x}^{\infty} \sum_{k=t}^{\infty} = \sum_{k=x}^{\infty} \sum_{t=x}^k$:

$$= \sum_{k=0}^{\infty} \alpha \delta_k^s \sum_{t=0}^k \nabla \hat{q}(S_t, A_t, \mathbf{w}_t) \prod_{i=t+1}^k \gamma_i \lambda_i \rho_i$$

This is in the form of a backward-view TD update if the inner sum can be written and updated incrementally as an eligibility trace. We now show this is possible. Define:

$$\mathbf{z}_k \doteq \sum_{t=0}^k \nabla \hat{q}(S_t, A_t, \mathbf{w}_t) \prod_{i=t+1}^k \gamma_i \lambda_i \rho_i$$

We can verify this can be updated incrementally from $\mathbf{z}_{k-1}$:

$$\begin{aligned}
\mathbf{z}_k &= \sum_{t=0}^k \nabla \hat{q}(S_t, A_t, \mathbf{w}_t) \prod_{i=t+1}^k \gamma_i \lambda_i \rho_i \\
&= \sum_{t=0}^{k-1} \nabla \hat{q}(S_t, A_t, \mathbf{w}_t) \prod_{i=t+1}^k \gamma_i \lambda_i \rho_i + \nabla \hat{q}(S_k, A_k, \mathbf{w}_k) \\
&= \gamma_k \lambda_k \rho_k \sum_{t=0}^{k-1} \nabla \hat{q}(S_t, A_t, \mathbf{w}_t) \prod_{i=t+1}^{k-1} \gamma_i \lambda_i \rho_i + \nabla \hat{q}(S_k, A_k, \mathbf{w}_k) \\
&= \gamma_k \lambda_k \rho_k \mathbf{z}_{k-1} + \nabla \hat{q}(S_k, A_k, \mathbf{w}_k)
\end{aligned}$$

Changing the index from k to t , this is exactly the eligibility trace update (12.29):

$$\boxed{\mathbf{z}_t \doteq \gamma_t \lambda_t \rho_t \mathbf{z}_{t-1} + \nabla \hat{q}(S_t, A_t, \mathbf{w}_t)}$$

which is the general accumulating trace update for action values, directly analogous to the state-value trace (12.25), with $\nabla \hat{v}(S_t, \mathbf{w}_t)$ replaced by $\nabla \hat{q}(S_t, A_t, \mathbf{w}_t)$ throughout. $\square$

Exercise 12.13

Question

What are the dutch-trace and replacing-trace versions of off-policy eligibility traces for state-value and action-value methods?

Answer

We take the general accumulating trace updates (12.25) and (12.29) and substitute in the Dutch trace and replacing trace definitions.

State-Value Methods

The general accumulating trace update (12.25) is:

$$\mathbf{z}_t \doteq \rho_t (\gamma_t \lambda_t \mathbf{z}_{t-1} + \nabla \hat{v}(S_t, \mathbf{w}_t))$$

Replacing trace (state-value):

$$z_{t,i} \doteq \rho_t \left(\gamma_t \lambda_t z_{t-1,i} + \begin{cases} 1 & \text{if } x_{t,i} = 1 \\ 0 & \text{otherwise} \end{cases} \right)$$

where $x_{t,i}$ is the i -th component of the feature vector at time t .

Dutch trace (state-value):

$$\mathbf{z}_t \doteq \rho_t (\gamma_t \lambda_t \mathbf{z}_{t-1} + (1 - \alpha \gamma_t \lambda_t \rho_t \mathbf{z}_{t-1}^\top \nabla \hat{v}(S_t, \mathbf{w}_t)) \nabla \hat{v}(S_t, \mathbf{w}_t))$$

Action-Value Methods

The general accumulating trace update (12.29) is:

$$\mathbf{z}_t \doteq \gamma_t \lambda_t \rho_t \mathbf{z}_{t-1} + \nabla \hat{q}(S_t, A_t, \mathbf{w}_t)$$

Replacing trace (action-value):

$$z_{t,i} \doteq \gamma_t \lambda_t \rho_t z_{t-1,i} + \begin{cases} 1 & \text{if } x_{t,i} = 1 \\ 0 & \text{otherwise} \end{cases}$$

Dutch trace (action-value):

$$\mathbf{z}_t \doteq \gamma_t \lambda_t \rho_t \mathbf{z}_{t-1} + (1 - \alpha \gamma_t \lambda_t \rho_t \mathbf{z}_{t-1}^\top \nabla \hat{q}(S_t, A_t, \mathbf{w}_t)) \nabla \hat{q}(S_t, A_t, \mathbf{w}_t)$$

In all cases the substitution is straightforward: the Dutch and replacing trace variants modify how the current gradient term is added to the decayed previous trace, while the off-policy correction via ρ_t carries through identically from the accumulating case. For action-value methods, $\nabla \hat{v}(S_t, \mathbf{w}_t)$ is replaced throughout by $\nabla \hat{q}(S_t, A_t, \mathbf{w}_t)$. $\square$